

Flight School Admin Dashboard — Technical & Functional Audit

SECTION 06-operations-it TYPE audit STATUS **active** OWNER GTM & CS Specialist

UPDATED 2026-08-19

[!NOTE] **Superseded 2026-08-19 — retained as a record, not corrected.** This document captured the position as it stood on its own date. The price card, the B2B pricing model and the platform architecture have all changed since: see [01-governance/decision-log.md](#) DEC-011 and [03-finance/monetization.md](#) for what is current. Nothing below has been edited — its value is that it records what was decided or observed at the time, including the parts that later turned out to be wrong.

1. Executive Summary & Audit Scope

This document provides a comprehensive audit of the **Flight School Admin & Instructor Dashboard** requirements, bridging the commercial GTM commitments made in [b2b-pipeline.md](#) and [gtm-schools.md](#) with the architectural specification in [spec-instructor-dashboard.md](#).

Key Audit Findings:

- Commercial Alignment:** The dashboard feature set directly fulfills all promises made on [schools.html](#) and B2B pilot sales decks (Cohort Exam Readiness %, Cadet Drill-Down, Weak Area Identification, Assigned Reading Paths).
- Architecture & Multi-Tenancy:** The sub-tree Firestore data model (`schools/{schoolId}/roster/{cadetUid}`) effectively preserves 100% per-user isolation of `users/{uid}` logbooks while enabling safe cross-user progress reads.
- Primary Gap Identified:** Study progress currently resides in client-side `localStorage` (`study-progress.js`). Phase A migration to Firestore (`users/{uid}/progress/summary`) is a strict prerequisite before any B2B pilot launch.

2. Multi-Tenant Architecture & Data Isolation Audit

Data Model Architecture

```
schools/{schoolId}
├── adminUids: string[]
├── instructorUids: string[]
├── seatLimit: number
└── roster/{cadetUid}
    ├── status: 'invited' | 'active' | 'removed'
    ├── consent: boolean
    ├── readiness: number (0-100)
    ├── lastMockScore: number
    ├── lessonsDone: number
    └── assignedPaths: string[]
```

Security Rules Isolation Audit

- **Isolation Principle:** Instructors **never** read into `users/{cadetUid}`. They read only `schools/{schoolId}/roster/{cadetUid}`.
- **Diff Check Enforcement:** Security rules enforce strict field diffing:
 - Cadets can only edit `readiness`, `lastMockScore`, `lessonsDone`, `currencyFlags`, and `progressUpdatedAt`.
 - Cadets can only edit if `roster().consent = true`.
 - Instructors can only edit `assignedPaths`.
 - Seat allocations (`seatLimit`, `seatsUsed`) and roster lifecycle are strictly Cloud Function Admin SDK operations.
- **Audit Assessment: PASS.** The design satisfies security and multi-tenancy rules without exposing user private collections.

3. Data Privacy & KSA PDPL Compliance Audit

Compliance Criteria	Specification Provision	Audit Status	Recommendation
Consent Management	Explicit opt-in modal on first cadet sign-in (<code>consent: true</code>)	PASS	Require re-consent upon annual contract renewal.
Scope Limitation	Only study progress and readiness scores shared; logbooks remain strictly private	PASS	Display explicit privacy badge ("Logbook Private") in cadet view.
Right to Revoke	Cadet can flip <code>consent: false</code> from Settings, immediately blanking shared roster entry	PASS	Ensure Cloud Function clears denormalized progress data on revocation.

Compliance Criteria	Specification Provision	Audit Status	Recommendation
DPIA Inclusion	Cross-user data sharing in B2B context	CONDITIONAL	Must update PDPL Data Protection Impact Assessment (DPIA) before pilot launch.

4. Functional Requirements Audit

Core Feature Coverage Matrix

Feature	Pitch / GTM Requirement	Technical Spec Coverage	Readiness
Cohort Readiness Rollup	Display average exam-readiness % across entire cohort	Calculated from <code>roster/{cadetUid}.readiness</code> average	Ready for build
Cadet Drill-Down	Drill into individual cadet mock scores & weak topics	Supported in <code>instructor.html</code> design	Ready for build
Weak Topic Filtering	Filter cadets by GACAR Part or topic deficiency	Filter UI supported in spec	Needs topic tag indexing
Reading Path Assignment	Instructors assign GACAR study modules (<code>assignedPaths</code>)	Supported via <code>roster/{cadetUid}.assignedPaths</code> array	Ready for build
Seat & Roster Management	Add/remove cadets, track seat limits	Server-side Cloud Functions (<code>grantSchoolLicence</code>)	Core logic built in <code>functions/school.js</code>

5. Technical Gap Analysis & Prerequisites

[Current State]	[Required Phase A]	[
Target B2B State]		
localStorage:	Firestore Document:	
Multi-Tenant Roster:		
study-progress.js	→ LocalStorage	→ users/{uid}/progress/summary
schools/{id}/roster/{uid}		→

1. Gap 1: Firestore Progress Summary (Phase A)

- *Issue:* `study-progress.js` currently stores mock exam scores and lesson completion exclusively in browser `localStorage`.

- *Remediation:* Build `users/{uid}/progress/summary` sync mechanism on client study session completion.

2. Gap 2: School Account Provisioning Tool (Phase C)

- *Issue:* Missing admin UI/CLI for provisioning school accounts (`createSchoolAccount`).
- *Remediation:* Create an internal Firebase Admin script/callable function for CS operations.

3. Gap 3: Executive Reporting Export

- *Issue:* Heads of Training require monthly PDF readiness summaries for regulatory audits.
- *Remediation:* Add a CSV/PDF export button in `instructor.html` generating cohort performance summaries.

6. Action Plan & Build Roadmap

Phase	Milestone	Priority	Effort	Target Completion
Phase A	Progress Persistence (<code>users/{uid}/progress/summary</code>)	P0 (Blocker)	3 Days	Week 1
Phase B	Firestore Security Rules & Emulator Unit Tests	P0 (Blocker)	4 Days	Week 2
Phase C	Cloud Functions (<code>grantSchoolLicence</code> , <code>createSchool</code> , <code>consent</code>)	P0 (Blocker)	4 Days	Week 2
Phase D	Instructor UI (<code>instructor.html</code> - Cohort View & Drill-Down)	P1	5 Days	Week 3
Phase E	Admin Roster Management & Reading Path Assignment	P1	4 Days	Week 4

Related Documents

- [Instructor Dashboard Design Spec](#)
- [B2B Pipeline Playbook](#)
- [GTM Flight Schools Strategy](#)
- [Customer Health Scoring Spec](#)